

SHASHANK S

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2+ years of experience designing, building, and scaling **Go (Golang) microservices** for production healthcare systems. Skilled in **distributed systems**, **event-driven architecture** (**Apache Kafka**), **gRPC**, **REST APIs**, **MongoDB**, **PostgreSQL**, **Docker**, and **CI/CD**. Reduced API latency by 85%+ and shipped an ABDM-certified Healthcare Information System to production.

WORK EXPERIENCE

Probeplus

Engineer I

Nov 2024 – Present

Bengaluru, Karnataka

- **Spearheaded** end-to-end design and development of **CareKonnnect**, an ABDM-certified Healthcare Information System, within a cross-functional **Agile** team, driving **M1-M4 compliance** through the complete **software development life cycle (SDLC)** from requirements to production deployment.
- **Architected** a scalable backend of **14+ microservices** using **Go (Fiber)** and **gRPC**, applying **object-oriented design patterns** (Factory, Strategy, Repository) to enforce domain-driven service boundaries and inter-service **Protobuf** contracts with high-cohesion system design.
- Engineered high-concurrency backend modules for **Queue Management** and **Automated Billing** behind **KrakenD API Gateway**, leveraging **Go goroutines**, **channels**, and **worker pools** (multi-threading and concurrency) for parallel request processing; maintained peak-load stability via pprof profiling and rate limiting.
- Implemented **HL7 FHIR** standards and full **ABHA** national health network integration; designed RESTful and event-driven APIs enabling seamless healthcare data exchange and national-level interoperability across distributed backend services.
- Deployed **event-driven architecture** using **Apache Kafka**, **MongoDB (NoSQL)**, and **Keycloak (IAM)**, ensuring robust security, real-time medical record processing, and end-to-end system observability with **Prometheus** and **Grafana**.
- Reduced API latency by **85%+** through systematic **root-cause analysis and performance tuning**: MongoDB compound indexes, query refactoring, connection pooling, and microservice caching, demonstrating strong debugging and problem-determination skills under production constraints.

MunarkTech India Pvt Ltd

Application Development Intern

Feb 2024 – Apr 2024

Remote, Karnataka

- Contributed to backend feature development and **RESTful API** integrations within an **Agile** sprint workflow, participating in code reviews, defect identification, and QA test plan definition across the full **SDLC**; gained hands-on exposure to application architecture and **version control (Git)** in a collaborative team environment.

PROJECTS

ImageCompressor-DCT | [Link](#) | Go, DCT, Huffman Coding, REST API, Docker

2026

- Implemented DCT-based image compression from scratch in **Go** (separable 2D transform, JPEG quantization, Huffman coding), achieving **48.3x** compression ratio at **PSNR 33.77 dB** and **SSIM 0.9924** on 1920x1080 images, demonstrating strong CS fundamentals in **algorithms and data structures**.
- Parallelised block-level DCT and quantization using **goroutine worker pools** (**multi-threading and concurrency**); precomputed cosine lookup table yields **2.4x** speedup, reducing encode time from 800 ms to **40 ms** on 1080p, verified via **REST** benchmark endpoint against Go stdlib JPEG, Python PIL, and OpenCV.

Yogavana Ayurpharmacy - E-Commerce Platform | [Link](#) | FastAPI, AWS S3, Razorpay

2025

- Designed and shipped a **production e-commerce platform** with **FastAPI** and **MongoDB**, covering catalog, cart, inventory, appointment booking, and **Razorpay** payment integration; **deployed live** and serving real users.
- Built **RESTful APIs** consumed by a **Next.js** frontend with **AWS S3** media storage; automated containerised build, test, and deployment via **Docker** and **Jenkins CI/CD**.

- Built packet capture tool from scratch using Linux **AF_PACKET raw sockets**, decoding Ethernet, IPv4, TCP, and UDP headers with **less than 0.1% packet loss** at 1 Gbps; displays real-time protocol distribution, bandwidth metrics, and top talkers.
- Implemented concurrent packet processing with **goroutine worker pools**: dedicated capture goroutine, parallel parsers, and async display; tracks active TCP flows and detects HTTP traffic patterns.
- Achieved **335 ns/packet** full-decode latency via `sync.Pool` (22% faster, 31% fewer allocations) and zero-alloc ring buffer; **2.9M pps** theoretical throughput gives **35x** headroom over a saturated 1 Gbps link.

TECHNICAL SKILLS

Languages: Go (Golang), Python, Java, SQL, Oracle SQL

Frameworks & Technologies: Go Fiber, gRPC, FastAPI, Spring Boot, Apache Kafka, REST APIs, Keycloak, KrakenD API Gateway

Protocols & Standards: HL7 FHIR, ABHA (ABDM), Protobuf, HTTP/2, REST, gRPC, Kafka Transactions (EOS)

Databases: MongoDB (NoSQL), PostgreSQL, Oracle SQL (indexing, query optimization, stored procedures)

Cloud & Infrastructure: Docker, Kubernetes (basics), Jenkins (CI/CD), AWS S3, Linux, Prometheus, Grafana

Engineering Practices: Microservices, Distributed Systems, Event-Driven Architecture, System Design, OOP Design Patterns, Agile/Scrum, SDLC, Multi-threading & Concurrency, Open-Source Development

Developer Tools: Git, GitHub, Bitbucket, Jira, Zoho, IntelliJ, VS Code

Open Source: Kafka Event-Streaming Patterns ([GitHub](#)) – Go reference library of 11 Apache Kafka patterns using IBM/sarama: DLQ with retry, idempotent producer, exactly-once semantics

EDUCATION

Dayananda Sagar University

B.Tech, Computer Science and Engineering (CGPA: 7.63)

Aug 2020 – Jun 2024

Bengaluru, Karnataka